DEVELOPMENT OF A LOW-COST FEEDBACK SYSTEM FOR POSTURE DETECTION

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Abstract

This project presents the design and implementation of a low-cost, real-time posture monitoring and feedback system designed to detect cervicothoracic kyphosis and forward head posture. The system utilizes an Arduino Uno microcontroller and an MPU6050 inertial measurement unit (IMU) mounted at the C7–T1 vertebral junction to measure angular deviations of the upper spine. A temporal filtering mechanism is implemented to distinguish sustained poor posture from brief movements. When an abnormal posture persists for more than five seconds, corrective feedback is provided through a servo motor, buzzer, and LED. The proposed system offers an affordable, non-invasive solution for continuous posture monitoring and corrective assistance.

Introduction

Poor posture, particularly cervicothoracic kyphosis, has become increasingly prevalent due to prolonged screen usage and sedentary lifestyles. Forward head posture places excessive mechanical stress on the cervical spine, leading to chronic pain and musculoskeletal disorders. Existing clinical posture assessment tools are expensive and lack real-time corrective feedback. This project focuses on developing a wearable, low-cost posture detection system that continuously monitors spinal alignment and provides immediate corrective cues to promote healthy posture habits.

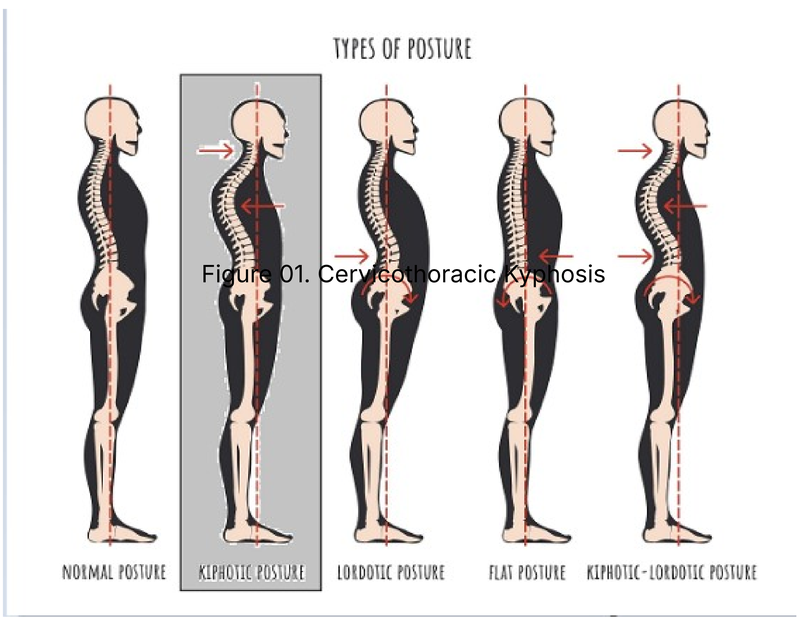


Figure 01. Cervicothoracic Kyphosis

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Methodology

- **Posture Sensing:** An MPU6050 IMU sensor, which combines a 3-axis accelerometer and a 3-axis gyroscope, is mounted at the C7–T1 vertebral junction to capture spinal orientation.
- **Microprocessor Processing**: The Arduino Uno receives sensor data via I²C communication and computes the forward bending angle relative to an upright reference posture.
- **Detection Logic:** A posture deviation threshold of 30° is defined. To avoid false triggering due to short movements, a 5-second temporal window is applied before confirming poor posture.
- **Feedback Mechanism**: Once sustained poor posture is detected. LED provides visual indication . Buzzer generates an audible alert. Servo motor produces a physical corrective cue

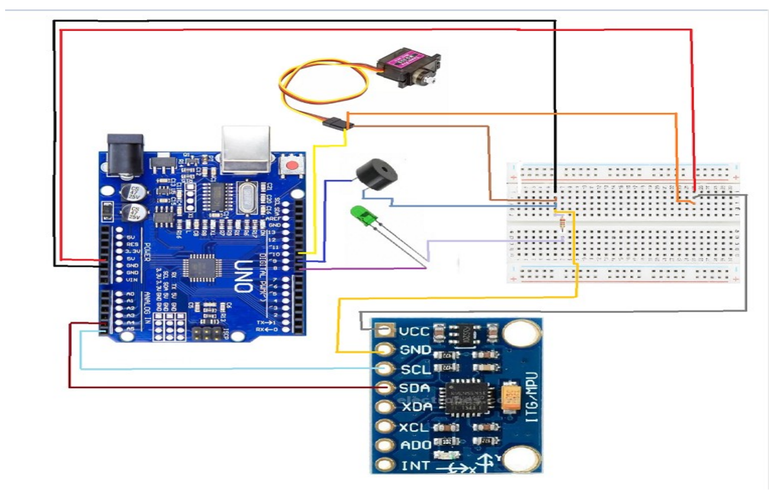


Figure 02. Circuit Diagram

Result

- The developed low-cost posture detection system was tested under normal daily activities and controlled posture variations. The system consistently detected sustained forward bending of the upper spine when the posture deviation exceeded the predefined threshold of 30 degrees for a continuous duration of 5 seconds. Short-term movements such as nodding, stretching, or minor posture adjustments did not activate the alert system, confirming the effectiveness of the temporal filtering logic in minimizing false alarms.
- Placement of the MPU6050 sensor at the C7–T1 vertebral junction provided high sensitivity to cervicothoracic posture changes, allowing accurate detection of forward head posture and kyphotic deviations. The feedback mechanisms operated reliably, where the LED provided instant visual confirmation, the buzzer generated an audible warning, and the servo motor delivered a clear mechanical cue. The system remained stable during extended operation and demonstrated consistent performance without unintended triggering.

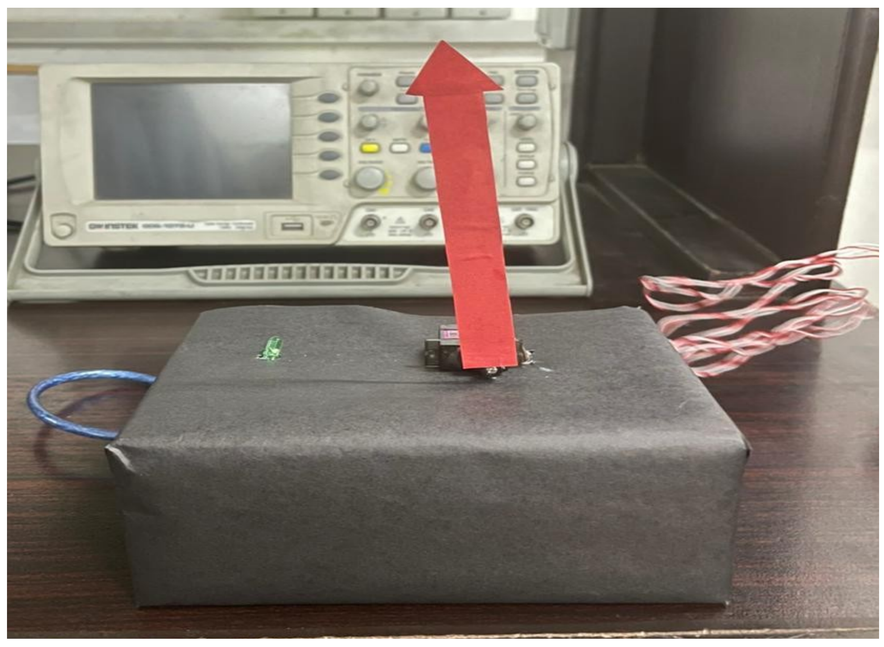


Figure 03. Project Model

Conclusion

This project successfully demonstrates the design and implementation of an affordable, real-time posture monitoring and feedback system for detecting cervicothoracic kyphosis. By integrating an IMU-based sensing approach with microprocessor-controlled decision logic, the system effectively monitors spinal alignment and provides immediate corrective feedback without the need for expensive clinical equipment. The incorporation of a 5-second temporal delay significantly improves system reliability by distinguishing harmful sustained poor posture from normal transient movements. The multi-modal feedback approach—visual, auditory, and mechanical enhances user awareness and encourages timely posture correction. Overall, the proposed system offers a practical and scalable solution for posture monitoring, with strong potential for future development in preventive healthcare, rehabilitation, and wearable biomedical applications.

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References

- *Real-Time Forward Head Posture Detection and Correction System Utilizing an Inertial Measurement Unit Sensor | MDPI*